高等量子力学　第 3 次作业

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题目十六　考虑外场作用下的一维谐振子系统：

$$\hat{H} = \hat{H}_0 + \hat{V}, \tag{1}$$

其中

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2}, \qquad \hat{V} = -\lambda\hat{x}. \tag{2}$$

设初始体系处于相干态上，求 t 时刻的平均值

$$\overline{x}_t = {}_I\langle \alpha, t \,|\, \hat{x}_I(t) \,|\, \alpha, t\rangle_I \tag{3}$$

（准确至 λ^2 量级）。

回答　对于自由谐振子，其哈密顿量 $\hat{H}_0$ 可写为

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{m\omega^2\hat{x}^2}{2} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + 1/2\right), \tag{4}$$

其中

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{\mathrm{i}}{m\omega}\hat{p}\right), \qquad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{\mathrm{i}}{m\omega}\hat{p}\right). \tag{5}$$

可以反解出

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a} + \hat{a}^\dagger\right). \tag{6}$$

定义

$$\hat{U}_0(t, t_0) = \mathrm{e}^{-\mathrm{i}\hat{H}_0(t-t_0)/\hbar}, \tag{7}$$

则相互作用绘景中的算符 $\hat{A}_I(t)$ 定义为

$$\hat{A}_I(t) = \hat{U}_0^\dagger(t, t_0)\hat{A}_S(t)\hat{U}_0(t, t_0). \tag{8}$$

相互作用绘景下，算符 $\hat{A}_I(t)$ 的时间演化满足如下的微分方程：

$$\frac{\mathrm{d}\hat{A}_I(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\left[\hat{A}_I(t), \hat{H}_0\right] + \left(\frac{\partial\hat{A}_S(t)}{\partial t}\right)_I. \tag{9}$$

由 $\left[\hat{H}_0, \hat{U}_0(t, t_0)\right] = 0$，分别计算：

$$\begin{aligned}
\left[\hat{H}_0, \hat{a}_I(t)\right] &= \left[\hat{H}_0, \hat{U}_0^\dagger(t, t_0)\hat{a}\hat{U}_0(t, t_0)\right] \\
&= \hat{H}_0\hat{U}_0^\dagger(t, t_0)\hat{a}\hat{U}_0(t, t_0) - \hat{U}_0^\dagger(t, t_0)\hat{a}\hat{U}_0(t, t_0)\hat{H}_0 \\
&= \hat{U}_0^\dagger(t, t_0)\hat{H}_0\hat{a}\hat{U}_0(t, t_0) - \hat{U}_0^\dagger(t, t_0)\hat{a}\hat{H}_0\hat{U}_0(t, t_0) \\
&= \hat{U}_0^\dagger(t, t_0)\left[\hat{H}_0, \hat{a}\right]\hat{U}_0(t, t_0) = \hat{U}_0^\dagger(t, t_0)\left[\hbar\omega\left(\hat{a}^\dagger\hat{a} + 1/2\right), \hat{a}\right]\hat{U}_0(t, t_0) \\
&= -\hbar\omega\hat{U}_0^\dagger(t, t_0)\hat{a}\hat{U}_0(t, t_0) \\
&= -\hbar\omega\hat{a}_I(t),
\end{aligned} \tag{10}$$

$$\begin{aligned}\left[\hat{H}_0, \hat{a}_I^\dagger(t)\right] &= \hat{U}_0^\dagger(t,t_0)\left[\hat{H}_0, \hat{a}^\dagger\right]\hat{U}_0(t,t_0) = \hat{U}_0^\dagger(t,t_0)\left[\hbar\omega\left(\hat{a}^\dagger\hat{a}+1/2\right), \hat{a}^\dagger\right]\hat{U}_0(t,t_0)\\ &= \hbar\omega\hat{U}_0^\dagger(t,t_0)\hat{a}^\dagger\hat{U}_0(t,t_0)\\ &= \hbar\omega\hat{a}_I^\dagger(t).\end{aligned} \tag{11}$$

因此，$\hat{a}_I(t), \hat{a}_I^\dagger(t)$ 分别满足：

$$\frac{\mathrm{d}\hat{a}_I(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\left[\hat{a}_I(t), \hat{H}_0\right] = -\mathrm{i}\omega\hat{a}_I(t), \tag{12}$$

$$\frac{\mathrm{d}\hat{a}_I^\dagger(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\left[\hat{a}_I^\dagger(t), \hat{H}_0\right] = \mathrm{i}\omega\hat{a}_I^\dagger(t). \tag{13}$$

设初始时刻 $t_0 = 0$，则结合初始条件 $\hat{a}_I(t=0) = \hat{a}, \hat{a}_I^\dagger(t=0) = \hat{a}^\dagger$ 可得：

$$\hat{a}_I(t) = \mathrm{e}^{-\mathrm{i}\omega t}\hat{a}, \quad \hat{a}_I^\dagger(t) = \mathrm{e}^{\mathrm{i}\omega t}\hat{a}^\dagger. \tag{14}$$

于是相互作用绘景中的坐标算符为

$$\hat{x}_I(t) = \sqrt{\frac{\hbar}{2m\omega}}\left(\hat{a}_I(t) + \hat{a}_I^\dagger(t)\right) = \sqrt{\frac{\hbar}{2m\omega}}\left(\mathrm{e}^{-\mathrm{i}\omega t}\hat{a} + \mathrm{e}^{\mathrm{i}\omega t}\hat{a}^\dagger\right). \tag{15}$$

相互作用绘景中的微扰算符为

$$\hat{V}_I(t) = -\lambda\hat{x}_I(t) = -\lambda\sqrt{\frac{\hbar}{2m\omega}}\left(\mathrm{e}^{-\mathrm{i}\omega t}\hat{a} + \mathrm{e}^{\mathrm{i}\omega t}\hat{a}^\dagger\right). \tag{16}$$

相互作用绘景中的态矢的定义为

$$|\psi_I(t)\rangle = \hat{U}_0^\dagger(t,t_0)\,|\psi_S(t)\rangle\,, \tag{17}$$

取 $t = t_0$ 可得两个绘景中初态的关系：

$$|\psi_I(t_0)\rangle = |\psi_S(t_0)\rangle\,. \tag{18}$$

相互作用绘景中态的演化形式上可写为

$$|\psi_I(t)\rangle = \hat{U}_I(t,t_0)\,|\psi_I(t_0)\rangle\,, \qquad \hat{U}_I(t,t_0) = \mathcal{T}\exp\left(-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\hat{V}_I(t')\right), \tag{19}$$

其中 $\mathcal{T}$ 是编时算符，它把时间参数大的算符放左边。形式上 $|\alpha,t\rangle_I$ 可写为

$$|\alpha,t\rangle = \hat{U}_I(t,0)\,|\alpha\rangle\,, \tag{20}$$

t 时刻坐标算符的期望值为：

$$\bar{x}_t = {}_I\langle\alpha,t\,|\,\hat{x}_I(t)\,|\,\alpha,t\rangle_I = \left\langle\alpha\,\middle|\,\hat{U}_I^\dagger(t,0)\hat{x}_I(t)\hat{U}_I(t,0)\,\middle|\,\alpha\right\rangle. \tag{21}$$

把相互作用绘景时间演化算符展开到二阶：

$$\begin{aligned}\hat{U}_I(t,0) &= \mathcal{T}\exp\left(-\frac{\mathrm{i}}{\hbar}\int_{t_0}^{t}\mathrm{d}t'\hat{V}_I(t')\right)\\ &\approx \hat{I} + \left(-\frac{\mathrm{i}}{\hbar}\int_0^t\mathrm{d}t_1\hat{V}_I(t_1)\right) + \frac{1}{2!}\left(-\frac{\mathrm{i}}{\hbar}\right)^2\int_0^t\mathrm{d}t_1\int_0^t\mathrm{d}t_2\mathcal{T}\left[\hat{V}_I(t_1)\hat{V}_I(t_2)\right]\\ &\approx \hat{I} + \left(-\frac{\mathrm{i}}{\hbar}\int_0^t\mathrm{d}t_1\hat{V}_I(t_1)\right) + \left(-\frac{\mathrm{i}}{\hbar}\right)^2\int_0^t\mathrm{d}t_1\int_0^{t_1}\mathrm{d}t_2\hat{V}_I(t_1)\hat{V}_I(t_2)\\ &\equiv \hat{U}_I^{(0)}(t,0) + \hat{U}_I^{(1)}(t,0) + \hat{U}_I^{(2)}(t,0),\end{aligned} \tag{22}$$

注意到

$$\begin{aligned}\bar{x}_t &= \left\langle \alpha \left| \hat{U}_I^\dagger(t,0)\hat{x}_I(t)\hat{U}_I(t,0) \right| \alpha \right\rangle \\ &\approx \left\langle \alpha \left| \left(\hat{U}_I^{(0)\dagger}(t,0)+\hat{U}_I^{(1)\dagger}(t,0)+\hat{U}_I^{(2)\dagger}(t,0)\right)\hat{x}_I(t)\left(\hat{U}_I^{(0)}(t,0)+\hat{U}_I^{(1)}(t,0)+\hat{U}_I^{(2)}(t,0)\right) \right| \alpha \right\rangle \\ &\approx \langle \hat{x}_I(t)\rangle_\alpha + \left\langle \hat{U}_I^{(1)\dagger}(t,0)\hat{x}_I(t)+\hat{x}_I(t)\hat{U}_I^{(1)}(t,0)\right\rangle_\alpha \\ &+ \left\langle \hat{U}_I^{(1)\dagger}(t,0)\hat{x}_I(t)\hat{U}_I^{(1)}(t,0)+\hat{U}_I^{(2)\dagger}(t,0)\hat{x}_I(t)+\hat{x}_I(t)\hat{U}_I^{(2)}(t,0)\right\rangle_\alpha \\ &\equiv \bar{x}_t^{(0)}+\bar{x}_t^{(1)}+\bar{x}_t^{(2)}.\end{aligned} \tag{23}$$

其中

$$\begin{aligned}\bar{x}_t^{(0)} &= \langle \hat{x}_I\rangle_\alpha = \sqrt{\frac{\hbar}{2m\omega}}\left(\mathrm{e}^{-\mathrm{i}\omega t}\langle\alpha\,|\,\hat{a}\,|\,\alpha\rangle + \mathrm{e}^{\mathrm{i}\omega t}\left\langle\alpha\left|\,\hat{a}^\dagger\,\right|\alpha\right\rangle\right) \\ &= \sqrt{\frac{\hbar}{2m\omega}}\left(\mathrm{e}^{-\mathrm{i}\omega t}\alpha+\mathrm{e}^{\mathrm{i}\omega t}\alpha^*\right).\end{aligned} \tag{24}$$

利用

$$[\hat{x}_I(t_1),\hat{x}_I(t)] = \frac{\hbar}{2m\omega}\left[\mathrm{e}^{-\mathrm{i}\omega t_1}\hat{a}+\mathrm{e}^{\mathrm{i}\omega t_1}\hat{a}^\dagger, \mathrm{e}^{-\mathrm{i}\omega t}\hat{a}+\mathrm{e}^{\mathrm{i}\omega t}\hat{a}^\dagger\right] = \frac{\mathrm{i}\hbar}{m\omega}\sin\left[\omega\left(t-t_1\right)\right], \tag{25}$$

有：

$$\begin{aligned}\bar{x}_t^{(1)} &= \left\langle \hat{U}_I^{(1)\dagger}(t,0)\hat{x}_I(t)+\hat{x}_I(t)\hat{U}_I^{(1)}(t,0)\right\rangle_\alpha = \frac{\mathrm{i}}{\hbar}\int_0^t \mathrm{d}t_1\left\langle\left[\hat{V}_I(t_1),\hat{x}_I(t)\right]\right\rangle_\alpha \\ &= (-\lambda)\frac{\mathrm{i}}{\hbar}\int_0^t \mathrm{d}t_1\left\langle[\hat{x}_I(t_1),\hat{x}_I(t)]\right\rangle_\alpha = \frac{\lambda}{m\omega}\int_0^t \mathrm{d}t_1\sin\left[\omega\left(t-t_1\right)\right] \\ &= \frac{\lambda\left(1-\cos\omega t\right)}{m\omega^2}.\end{aligned} \tag{26}$$

利用

$$\begin{aligned}\hat{U}_I^{(1)\dagger}(t,0)\hat{x}_I(t)\hat{U}_I^{(1)}(t,0) &= \frac{1}{\hbar^2}\int_0^t \mathrm{d}t_1\int_0^t \mathrm{d}t_2\hat{V}_I(t_1)\hat{x}_I(t)\hat{V}_I(t_2) \\ &= \frac{1}{\hbar^2}\int_0^t \mathrm{d}t_1\int_0^{t_1} \mathrm{d}t_2\left[\hat{V}_I(t_1)\hat{x}_I(t)\hat{V}_I(t_2)+\hat{V}_I(t_2)\hat{x}_I(t)\hat{V}_I(t_1)\right]\end{aligned} \tag{27}$$

有：

$$\begin{aligned}\bar{x}_t^{(2)} &= \left\langle \hat{U}_I^{(1)\dagger}(t,0)\hat{x}_I(t)\hat{U}_I^{(1)}(t,0)+\hat{U}_I^{(2)\dagger}(t,0)\hat{x}_I(t)+\hat{x}_I(t)\hat{U}_I^{(2)}(t,0)\right\rangle_\alpha \\ &= \frac{1}{\hbar^2}\int_0^t \mathrm{d}t_1\int_0^{t_1} \mathrm{d}t_2\Big\langle\left[\hat{V}_I(t_1)\hat{x}_I(t)\hat{V}_I(t_2)+\hat{V}_I(t_2)\hat{x}_I(t)\hat{V}_I(t_1)\right. \\ &\left.-\hat{V}_I(t_2)\hat{V}_I(t_1)\hat{x}_I(t)-\hat{x}_I(t)\hat{V}_I(t_1)\hat{V}_I(t_2)\right]\Big\rangle_\alpha \\ &= -\frac{1}{\hbar^2}\int_0^t \mathrm{d}t_1\int_0^{t_1} \mathrm{d}t_2\left\langle\left[\hat{V}_I(t_2),\left[\hat{V}_I(t_1),\hat{x}_I(t)\right]\right]\right\rangle_\alpha \\ &= -\frac{\lambda^2}{\hbar^2}\int_0^t \mathrm{d}t_1\int_0^{t_1} \mathrm{d}t_2\left\langle[\hat{x}_I(t_2),[\hat{x}_I(t_1),\hat{x}_I(t)]]\right\rangle_\alpha \\ &= 0.\end{aligned} \tag{28}$$

综上，准确到二阶小量，

$$\bar{x}_t \approx \bar{x}_t^{(0)}+\bar{x}_t^{(1)}+\bar{x}_t^{(2)} = \sqrt{\frac{\hbar}{2m\omega}}\left(\mathrm{e}^{-\mathrm{i}\omega t}\alpha+\mathrm{e}^{\mathrm{i}\omega t}\alpha^*\right)+\frac{\lambda\left(1-\cos\omega t\right)}{m\omega^2}. \tag{29}$$

题目十七 从本征值方程

$$\hat{x}_H(t)\,|x,t\rangle_H = x\,|x,t\rangle_H \tag{30}$$

出发，导出自由粒子和谐振子系统中

$$K^*(x,t;x_0,t_0) = {}_H\langle x_0,t_0\,|\,x,t\rangle_H \tag{31}$$

满足的微分方程，进而导出传播子 $K(x,t;x_0,t_0)$ 的表达式。

回答 由海森堡绘景中算符、态矢的定义，有：

$$\hat{x}_H(t) = \hat{U}^\dagger(t,t_0)\hat{x}\hat{U}(t,t_0), \qquad |x,t\rangle_H = \hat{U}^\dagger(t,t_0)\,|x\rangle\,, \tag{32}$$

当 $t=t_0$ 时，有：

$$\hat{x}_H(t_0) = \hat{x}, \qquad |x,t_0\rangle_H = |x\rangle\,. \tag{33}$$

于是可以在海森堡绘景下表达传播子：

$$K(x,t;x_0,t_0) = \left\langle x\left|\,\hat{U}(t,t_0)\,\right|x_0\right\rangle = {}_H\langle x,t\,|\,x_0,t_0\rangle_H\,, \qquad K^*(x,t;x_0,t_0) = {}_H\langle x_0,t_0\,|\,x,t\rangle_H\,. \tag{34}$$

对于不显含时间的哈密顿量，时间演化算符形式上可写为

$$\hat{U}(t,t_0) = \mathrm{e}^{-\mathrm{i}\hat{H}(t-t_0)/\hbar}, \qquad |x,t\rangle_H = \hat{U}^\dagger(t,t_0)\,|x\rangle = \mathrm{e}^{\mathrm{i}\hat{H}(t-t_0)/\hbar}\,|x\rangle\,. \tag{35}$$

两边对 t 求导，得：

$$-\mathrm{i}\hbar\frac{\partial}{\partial t}\,|x,t\rangle_H = \hat{H}\,|x,t\rangle_H\,. \tag{36}$$

上式两边左乘 ${}_H\langle x_0,t_0|$ 得到传播子满足的微分方程：

$$-\mathrm{i}\hbar\frac{\partial}{\partial t}K^*(x,t;x_0,t_0) = {}_H\left\langle x_0,t_0\left|\,\hat{H}\,\right|x,t\right\rangle_H\,. \tag{37}$$

(a) 对于自由粒子，其哈密顿量为 $\hat{H}=\hat{p}^2/(2m)$。海森堡绘景中不显含时间的算符演化满足

$$\frac{\mathrm{d}\hat{A}_H(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\left[\hat{A}_H(t),\hat{H}\right]. \tag{38}$$

$$\frac{\mathrm{d}\hat{x}_H(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\left[\hat{x}_H(t),\hat{H}\right] = \frac{1}{\mathrm{i}\hbar}\hat{U}^\dagger(t,t_0)\left[\hat{x},\hat{H}\right]\hat{U}(t,t_0) = \hat{U}^\dagger(t,t_0)\frac{\hat{p}}{m}\hat{U}(t,t_0) = \frac{\hat{p}_H(t)}{m}, \tag{39}$$

$$\frac{\mathrm{d}\hat{p}_H(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\left[\hat{p}_H(t),\hat{H}\right] = \frac{1}{\mathrm{i}\hbar}\hat{U}^\dagger(t,t_0)\left[\hat{p},\hat{H}\right]\hat{U}(t,t_0) = 0, \tag{40}$$

结合初始条件 $\hat{p}_H(t_0)=\hat{p},\hat{x}_H(t_0)=\hat{x}$ 解得：

$$\hat{p}_H(t) = \hat{p}_H(t_0) = \hat{p}, \qquad \hat{x}_H(t) = \hat{x}_H(t_0) + \frac{t-t_0}{m}\hat{p}_H(t_0) = \hat{x} + \frac{t-t_0}{m}\hat{p}. \tag{41}$$

把上式代入本征方程 $\hat{x}_H(t)\,|x,t\rangle_H = x\,|x,t\rangle_H$，并左乘 ${}_H\langle x_0,t_0|$ 得到：

$$\left(x_0 - \frac{\mathrm{i}\hbar\,(t-t_0)}{m}\frac{\partial}{\partial x_0}\right)K^*(x,t;x_0,t_0) = xK^*(x,t;x_0,t_0). \tag{42}$$

可以解得：

$$K^*(x,t;x_0,t_0) = C_1(x,t;t_0)\exp\left[-\frac{\mathrm{i}m\,(x-x_0)^2}{2\hbar(t-t_0)}\right] \tag{43}$$

再计算矩阵元

$$\begin{aligned} {}_H\left\langle x_0,t_0\left|\hat{H}\right|x,t\right\rangle_H &= \left\langle x_0\left|\frac{\hat{p}^2}{2m}\right|x,t\right\rangle_H = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_0^2}\left\langle x_0\,|\,x,t\right\rangle_H \\ &= -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_0^2}{}_H\left\langle x_0,t_0\,|\,x,t\right\rangle_H = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_0^2}K^*(x,t;x_0,t_0), \end{aligned} \tag{44}$$

代入 (37) 得到方程：

$$-\mathrm{i}\hbar\frac{\partial}{\partial t}K^*(x,t;x_0,t_0) = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_0^2}K^*(x,t;x_0,t_0), \tag{45}$$

也即：

$$-\mathrm{i}\hbar\frac{\partial C_1(x,t;t_0)}{\partial t} = \frac{\mathrm{i}\hbar}{2(t-t_0)}C_1(x,t;t_0), \tag{46}$$

解得：

$$C_1(x,t;t_0) = C_2(x,t_0)\frac{1}{\sqrt{t-t_0}}, \tag{47}$$

$$K^*(x,t;x_0,t_0) = \frac{C_2(x,t_0)}{\sqrt{t-t_0}}\exp\left[-\frac{\mathrm{i}m\,(x-x_0)^2}{2\hbar(t-t_0)}\right]. \tag{48}$$

利用初始条件 $\lim\limits_{t\to t_0^+}K(x,t;x_0,t_0)=\delta(x-x_0)$ 以及公式

$$\delta(x-x_0) = \lim_{\tau\to 0^+}\sqrt{\frac{m}{-2\pi\mathrm{i}\hbar\tau}}\exp\left[-\frac{\mathrm{i}m(x-x_0)^2}{2\hbar\tau}\right], \tag{49}$$

对比可得：

$$C_2(x,t_0) = \sqrt{\frac{m}{-2\pi\mathrm{i}\hbar}}, \qquad K^*(x,t;x_0,t_0) = \sqrt{\frac{m}{-2\pi\mathrm{i}\hbar(t-t_0)}}\exp\left[-\frac{\mathrm{i}m\,(x-x_0)^2}{2\hbar(t-t_0)}\right], \tag{50}$$

$$K(x,t;x_0,t_0) = \sqrt{\frac{m}{2\pi\mathrm{i}\hbar(t-t_0)}}\exp\left[\frac{\mathrm{i}m\,(x-x_0)^2}{2\hbar(t-t_0)}\right]. \tag{51}$$

(b) 对于谐振子，哈密顿量为 $\hat{H}=\hat{p}^2/(2m)+m\omega^2\hat{x}^2/2$，不显含时间，因此

$$\frac{\mathrm{d}\hat{x}_H(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\left[\hat{x}_H(t),\hat{H}\right] = \frac{\hat{p}_H(t)}{m}, \tag{52}$$

$$\frac{\mathrm{d}\hat{p}_H(t)}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\left[\hat{p}_H(t),\hat{H}\right] = -m\omega^2\hat{x}_H(t), \tag{53}$$

联立得到

$$\ddot{\hat{x}}_H(t)+\omega^2\hat{x}_H(t)=0. \tag{54}$$

结合初始条件

$$\hat{x}_H(t_0)=\hat{x}, \qquad \hat{p}_H(t_0)=\hat{p}, \qquad \dot{\hat{x}}_H(t_0)=\frac{\hat{p}_H(t_0)}{m}=\frac{\hat{p}}{m}, \tag{55}$$

解得

$$\hat{x}_H(t) = \hat{x}\cos\left[\omega(t-t_0)\right]+\frac{\hat{p}}{m\omega}\sin\left[\omega(t-t_0)\right], \tag{56}$$

$$\hat{p}_H(t) = -m\omega\hat{x}\sin\left[\omega(t-t_0)\right]+\hat{p}\cos\left[\omega(t-t_0)\right]. \tag{57}$$

上式代入海森堡绘景坐标算符本征方程：

$$\left(\hat{x}\cos\left[\omega(t-t_0)\right]+\frac{\hat{p}}{m\omega}\sin\left[\omega(t-t_0)\right]\right)|x,t\rangle_H = x\,|x,t\rangle_H\,, \tag{58}$$

左乘 ${}_H\langle x_0,t_0|$ 得：

$$\left(x_0\cos\left[\omega(t-t_0)\right]-\frac{\mathrm{i}\hbar}{m\omega}\sin\left[\omega(t-t_0)\right]\frac{\partial}{\partial x_0}\right)K^*(x,t;x_0,t_0)=xK^*(x,t;x_0,t_0), \tag{59}$$

整理得到：

$$\frac{\partial K^*(x,t;x_0,t_0)}{\partial x_0}=\frac{\mathrm{i}m\omega}{\hbar}\left(x\csc\left[\omega(t-t_0)\right]-x_0\cot\left[\omega(t-t_0)\right]\right)K^*(x,t;x_0,t_0), \tag{60}$$

解得：

$$K^*(x,t;x_0,t_0)=C_1(x,t;t_0)\exp\left[\frac{\mathrm{i}m\omega}{\hbar}\left(-\frac{x_0^2\cot\left[\omega(t-t_0)\right]}{2}+xx_0\csc\left[\omega(t-t_0)\right]\right)\right], \tag{61}$$

再计算矩阵元

$$\begin{aligned}{}_H\left\langle x_0,t_0\left|\hat{H}\right|x,t\right\rangle_H &= \left\langle x_0\left|\frac{\hat{p}^2}{2m}+\frac{1}{2}m\omega^2\hat{x}^2\right|x,t\right\rangle_H = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_0^2}+\frac{m\omega^2x_0^2}{2}\right)\langle x_0\,|\,x,t\rangle_H\\ &=\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_0^2}+\frac{m\omega^2x_0^2}{2}\right){}_H\langle x_0,t_0\,|\,x,t\rangle_H\\ &=\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_0^2}+\frac{m\omega^2x_0^2}{2}\right)K^*(x,t;x_0,t_0),\end{aligned} \tag{62}$$

代入 (37) 得到谐振子传播子满足的微分方程

$$-\mathrm{i}\hbar\frac{\partial}{\partial t}K^*(x,t;x_0,t_0)=\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_0^2}+\frac{m\omega^2x_0^2}{2}\right)K^*(x,t;x_0,t_0). \tag{63}$$

整理得到：

$$-\mathrm{i}\hbar\frac{\partial C_1(x,t;t_0)}{\partial t}=\left(\frac{\mathrm{i}\hbar\omega}{2}\cot\left[\omega(t-t_0)\right]+\frac{m\omega^2x^2}{2}\csc^2\left[\omega(t-t_0)\right]\right)C_1(x,t;t_0), \tag{64}$$

利用积分公式

$$\int\cot(\omega\tau)\mathrm{d}\tau=\frac{1}{\omega}\ln|\sin(\omega\tau)|+C,\quad \int\csc^2(\omega\tau)\mathrm{d}\tau=-\frac{1}{\omega}\cot(\omega\tau)+C, \tag{65}$$

解得：

$$C_1(x,t;t_0)=\frac{C_2(x,t_0)}{\sqrt{\sin\left[\omega(t-t_0)\right]}}\exp\left[-\frac{\mathrm{i}m\omega x^2}{2\hbar}\cot\left[\omega(t-t_0)\right]\right], \tag{66}$$

$$K^*(x,t;x_0,t_0)=\frac{C_2(x,t_0)}{\sqrt{\sin\left[\omega(t-t_0)\right]}}\exp\left[-\frac{\mathrm{i}m\omega\left(\left(x^2+x_0^2\right)\cos\left[\omega(t-t_0)\right]-2xx_0\right)}{2\hbar\sin\left[\omega(t-t_0)\right]}\right], \tag{67}$$

考虑 $t\to t_0^+$，此时 $\sin\left[\omega(t-t_0)\right]\sim\omega(t-t_0),\cos\left[\omega(t-t_0)\right]\sim1$，于是

$$\lim_{t\to t_0^+}K^*(x,t;x_0,t_0)=\lim_{t\to t_0^+}\frac{C_2(x,t_0)}{\sqrt{\omega(t-t_0)}}\exp\left[-\frac{\mathrm{i}m\left(x-x_0\right)^2}{2\hbar(t-t_0)}\right] \tag{68}$$

由初始条件 $\lim\limits_{t\to t_0^+}K(x,t;x_0,t_0)=\delta(x-x_0)$ 以及 (49) 可得：

$$C_2(x,t_0)=\sqrt{\frac{m\omega}{-2\pi\mathrm{i}\hbar}}, \tag{69}$$

$$K(x,t;x_0,t_0)=\sqrt{\frac{m\omega}{2\pi\mathrm{i}\hbar\sin\left[\omega(t-t_0)\right]}}\exp\left[\frac{\mathrm{i}m\omega\left(\left(x^2+x_0^2\right)\cos\left[\omega(t-t_0)\right]-2xx_0\right)}{2\hbar\sin\left[\omega(t-t_0)\right]}\right]. \tag{70}$$

题目十八 设自由粒子的初始态为

$$\psi(x,t_0)=\frac{1}{(2\pi a^2)^{1/4}}\exp\left[-\frac{(x-x_0)^2}{4a^2}\right]. \tag{71}$$

试通过传播子求 t 时刻的波函数 $\psi(x,t)$，并讨论 $|\psi(x,t)|^2$ 在 t 值很大时的渐进行为。

回答 自由粒子传播子为

$$K(x,t;x',t_0)=\sqrt{\frac{m}{2\pi\mathrm{i}\hbar(t-t_0)}}\exp\left[\frac{\mathrm{i}m\left(x-x'\right)^2}{2\hbar(t-t_0)}\right], \tag{72}$$

t 时刻波函数为

$$\begin{aligned}\psi(x,t)&=\int_{-\infty}^{+\infty}K(x,t;x',t_0)\psi(x',t_0)\mathrm{d}x'\\&=\frac{1}{(2\pi a^2)^{1/4}}\sqrt{\frac{m}{2\pi\mathrm{i}\hbar(t-t_0)}}\int_{-\infty}^{+\infty}\exp\left[\frac{\mathrm{i}m\left(x-x'\right)^2}{2\hbar(t-t_0)}-\frac{(x'-x_0)^2}{4a^2}\right]\mathrm{d}x'\\&=\frac{1}{(2\pi a^2)^{1/4}}\sqrt{\frac{m}{2\pi\mathrm{i}\hbar(t-t_0)}}\int_{-\infty}^{+\infty}\exp\left[\left(-\frac{1}{4a^2}+\frac{\mathrm{i}m}{2\hbar(t-t_0)}\right)x'^2+\left(\frac{x_0}{2a^2}-\frac{\mathrm{i}mx}{\hbar(t-t_0)}\right)x'\right.\\&\left.+\frac{\mathrm{i}mx^2}{2\hbar(t-t_0)}-\frac{x_0^2}{4a^2}\right]\mathrm{d}x',\end{aligned} \tag{73}$$

定义无量纲时间参数

$$\eta=\frac{\hbar(t-t_0)}{2ma^2}, \tag{74}$$

$$\psi(x,t)=\frac{1}{(2\pi a^2)^{1/4}}\sqrt{\frac{1}{4\pi\mathrm{i}a^2\eta}}\exp\left(\frac{\mathrm{i}x^2-\eta x_0^2}{4a^2\eta}\right)\int_{-\infty}^{+\infty}\exp\left[\left(-\frac{\eta-\mathrm{i}}{4a^2\eta}\right)x'^2+\left(\frac{\eta x_0-\mathrm{i}x}{2a^2\eta}\right)x'\right]\mathrm{d}x'. \tag{75}$$

利用高斯积分公式

$$\int_{-\infty}^{+\infty}\mathrm{e}^{-Ax^2+Bx}\mathrm{d}x=\sqrt{\frac{\pi}{A}}\exp\left(\frac{B^2}{4A}\right),\quad A=\frac{\eta-\mathrm{i}}{4a^2\eta},\quad B=\frac{\eta x_0-\mathrm{i}x}{2a^2\eta}, \tag{76}$$

$$\psi(x,t)=\frac{1}{(2\pi a^2)^{1/4}}\frac{1}{\sqrt{1+\mathrm{i}\eta}}\exp\left[-\frac{(x-x_0)^2}{4a^2(1+\mathrm{i}\eta)}\right], \tag{77}$$

$$|\psi(x,t)|^2=\frac{1}{(2\pi a^2)^{1/2}}\frac{1}{\sqrt{1+\eta^2}}\exp\left[-\frac{(x-x_0)^2}{2\left(1+\eta^2\right)a^2}\right]. \tag{78}$$

当 $t\to+\infty$ 时，$\eta\to+\infty$，概率密度仍是高斯波包，波包中心不变，但展宽随时间线性增大。

题目十九 设系统经典拉氏量为

$$L=\frac{m\dot{x}^2}{2}-\lambda x. \tag{79}$$

求传播子 $K(x,t;x_0,t_0)$ 的表达式。

回答

题目二十 动量表象传播子 $k(p,t;p_0,t_0)$ 定义为

$$\varphi(p,t)=\int k(p,t;p_0,t_0)\varphi(p_0,t_0)\,\mathrm{d}p_0. \tag{80}$$

已知动量表象波函数 $\varphi(p,t)$ 与坐标表象波函数 $\psi(x,t)$ 间的变换关系为

$$\psi(x,t)=\frac{1}{\sqrt{2\pi\hbar}}\int \mathrm{e}^{\mathrm{i}px/\hbar}\varphi(p,t)\,\mathrm{d}p. \tag{81}$$

试求自由粒子系统的动量表象传播子。

回答